

How has bioinformatics been used to enhance our understanding on the causes of human diseases?

Protein Language Models Reveal the Functional Impact of CFTR Missense Variants: A Bioinformatics Case Study

Introduction

Bioinformatics is a relatively young but rapidly expanding field that integrates biological knowledge with computational and mathematical approaches to analyse molecular data. Its foundations were laid in the mid-20th century, when early protein sequencing and the discovery of DNA's structure created the need for systematic methods to store and compare sequences (Gauthier et al., 2019). As Sanger sequencing (Sanger, Nicklen and Coulson, 1977) and later high-throughput genomics accelerated data production, new computational tools became essential for organising and interpreting vast genomic and proteomic datasets. Early bioinformatics made use of classical machine-learning techniques such as regression, clustering, and decision trees to map gene–disease relationships and build functional annotations, but these methods struggled with increasingly complex “omics” data.

The emergence of deep learning marked a major shift by enabling models to capture nonlinear, high-dimensional patterns in biological data, leading to advances in protein structure prediction (Jumet al., 2021), variant effect estimation (Riesselman 2018), and disease mechanism modelling (Zhou et al, 2015). This essay examines how bioinformatics helps uncover the molecular causes of human disease through a case study of the *Cystic Fibrosis Transmembrane Conductance Regulator (CFTR) gene*. It further explores how modern deep-learning approaches—particularly transformer-based protein language models (PLMs)—can be applied to analyse pathogenic missense mutations. Using ESM-2, this study demonstrates how sequence-based models can reveal functional and mechanistic insights directly from amino-acid variation.

CFTR mutations and cystic fibrosis (CF)

CFTR functions as a phosphorylation-regulated, ATP-gated anion channel (Linsdell, 2018). CFTR's primary function is to maintain proper hydration and pH of epithelial surfaces in the lung, pancreas, intestine, and other organs (Csanády et al, 2019). Phosphorylation of the R-domain by PKA allows NBD1 and NBD2 to bind ATP and dimerise, a conformational switch that opens the transmembrane

pore for chloride and bicarbonate flow (See Fig. 1). Hydrolysis of ATP at NBD2 drives channel closure (Gadsby, Vergani and Csanády, 2006; Hwang and Kirk, 2013; Hanrahan and Wioland, 2004).

CF-causing mutations disrupt this cycle at multiple mechanistic levels (De Boeck and Amaral, 2016): Class I mutations prevent protein synthesis (nonsense mutations); Class II mutations (e.g., $\Delta F508$) impair folding and ER exit; Class III mutations (e.g., G551D) impair ATP-dependent gating; Class IV mutations reduce ionic conductance; Class V mutations reduce transcript abundance; and Class VI mutations destabilise the channel at the membrane. These defects reduce epithelial anion secretion and cause dehydrated airway mucus, ultimately leading to CF.

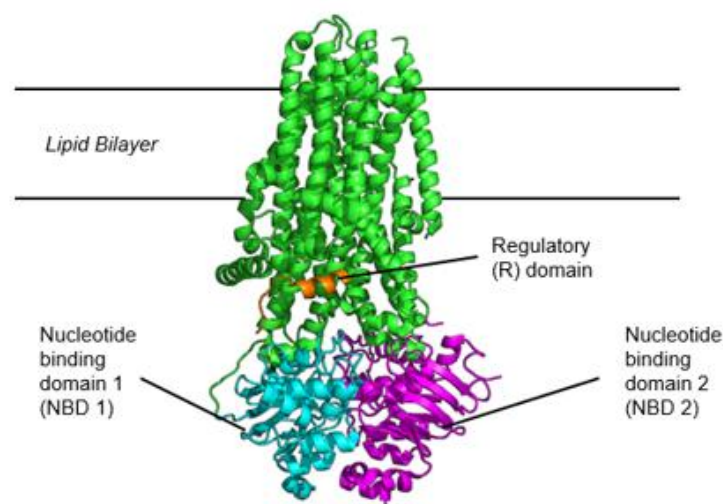


Figure 1. Structure of a human Cystic Fibrosis Transmembrane Conductance Regulator (CFTR).

Orange represents the regulatory domain (R domain) which gets phosphorylated. Teal represents the nucleotide binding domain 1 (NBD 1). Magenta represents NBD 2. Green represents transmembrane regions.

Bioinformatics and CFTR

After the discovery of the *CFTR* gene in 1989 by positional cloning, bioinformatics immediately provided clues to CFTR's physiological role and suggested concrete directions for functional work. Sequence comparisons of CFTR's nucleotide-binding domains (NBDs) revealed canonical Walker A, Walker B, and the LSGGQ "signature" motifs that define ABC ATP-binding domains, strongly implying that CFTR would be an ATP-regulated transporter or channel rather than a generic membrane protein (Riordan *et al.*, 1989; Lewis *et al.*, 2004; Kos and Ford, 2009). Comparative genomics across species then highlighted residues that are evolutionarily invariant and therefore likely to be functionally

critical. Many of these conserved sites later emerged as hotspots for disease-causing mutations: for example, G551 in the NBD1 signature motif is invariant in ABC NBDs, and clinical sequencing identified G551D as a recurrent CF allele (Chen *et al.*, 2001; Rishishwar *et al.*, 2012; Schneider and Hunke, 1998; Bompadre *et al.*, 2007; Wang *et al.*, 2009). Follow-up studies showed that G551D CFTR traffics to the membrane but essentially loses ATP-dependent gating, experimentally confirming the functional importance that bioinformatic analysis had predicted (Bompadre *et al.*, 2007; Wang *et al.*, 2020; Abu-Arish *et al.*, 2022). In this way, early sequence- and conservation-based bioinformatics did not just annotate CFTR; it *prioritised* specific residues and domains for *targeted* electrophysiological and structural studies.

Conversely, bioinformatics *integrates* decades of reductionist biochemistry into a coherent, systems-level view. By combining experimentally validated interactors—such as ENaC (Ismailov *et al.*, 1996), SNAREs (Cormet-Boyaka *et al.*, 2002), and PDZ scaffolds (Raghuram, Mak & Foskett, 2001)—with high-throughput affinity-purification mass spectrometry, researchers have computationally mapped the broader CFTR interactome (Pankow, S. *et al.*, 2016). Gene Ontology and pathway enrichment analyses then organise these proteins into functional modules like ‘ion transport’ and ‘trafficking’ (Iazzi *et al.*, 2022). This synthesis allows scientists to move beyond single-channel defects and investigate how the loss of CFTR perturbs *entire* cellular networks.

Deep Learning and the Emergence of PLMs

Traditional deep-learning models such as multilayer perceptrons (MLPs) and convolutional/recurrent neural networks helped bioinformatics move beyond classic statistical learning, but they struggled to capture long-range relationships within biological sequences. A major shift occurred with the development of the attention mechanism (Vaswani *et al.*, 2017), which allows a model to dynamically weight the importance of different positions in a sequence. This innovation led to the Transformer architecture, now the foundation of large language models like ChatGPT and Gemini. Because proteins are also linear sequences—*ordered amino acids* (AAs) rather than *words*—scientists began applying Transformers to biological sequence data, giving rise to PLMs (Rao *et al.*, 2019; Rives *et al.*, 2021; Elnaggar *et al.*, 2021; Vig *et al.*, 2021).

Meta AI’s ESM-1b (Rives *et al.*, 2021) and the more advanced ESM-2 (Lin *et al.*, 2023) are among the most widely used PLMs. Trained on millions of natural protein sequences, these pre-trained models learn the “grammar” of proteins and produce a high-dimensional vector embedding (e.g., 1280-dimensional vector) for every amino-acid position. These embeddings encode latent biochemical

information such as structural constraints and evolutionary conservation.

ESM models also use a masked-language modelling objective, where a residue is masked and the model predicts the probability distribution over all possible 20 AAs at that position (Rives *et al.*, 2021). This makes the model highly useful for studying missense mutations. For a mutation at position i , we obtain the predicted probability of the mutant AA $P(\text{mut}_i \mid \text{context})$ and compare it to that of the wild-type residue $P(\text{WT}_i \mid \text{context})$.

This simple comparison can be extended to calculating the log-likelihood ratio (LLR) which quantifies how compatible the mutation is with the learned protein grammar (Lin *et al.*, 2024):

$$\text{LLR}_i = \log P(\text{mut}_i \mid \text{context}) - \log P(\text{WT}_i \mid \text{context}).$$

A strongly negative LLR indicates that the model finds the mutant residue evolutionarily or biophysically implausible, suggesting potential functional disruption. In this essay, I applied ESM-2 to compute LLR scores for CFTR missense variants, demonstrating how modern deep-learning models can help identify disease-causing mutations from sequence alone.

Using ESM-2 for analysis of missense CFTR mutations

This analysis focused exclusively on missense CFTR mutations, as these variants can be directly evaluated using the log-likelihood ratio (LLR) framework established earlier. Because missense mutations involve the substitution of a single AA, ESM-2 can meaningfully compare the predicted probability of the wild-type residue with that of the mutant residue at the same sequence position. By contrast, Class I (nonsense/frameshift), Class V (splicing or regulatory) and Class VI (reduced stability) mutations are not predominantly simple single-residue substitutions and therefore cannot be interpreted within an LLR-based sequence model. For this reason, this study examined only Class II, Class III, and Class IV variants—the CFTR mutation classes that are predominantly missense.

A curated set of 14 benign, 11 Class II, 12 Class III, and 15 Class IV clinically validated missense variants were collected from the CFTR2 (Castellani *et al.*, 2013) and ClinVar (Landrum *et al.*, 2016) databases. LLR values for each mutation were obtained using ESM-2, and these distributions were visualised using a boxplot (Fig. 2) to assess whether the model captures biologically meaningful differences among mutation classes.

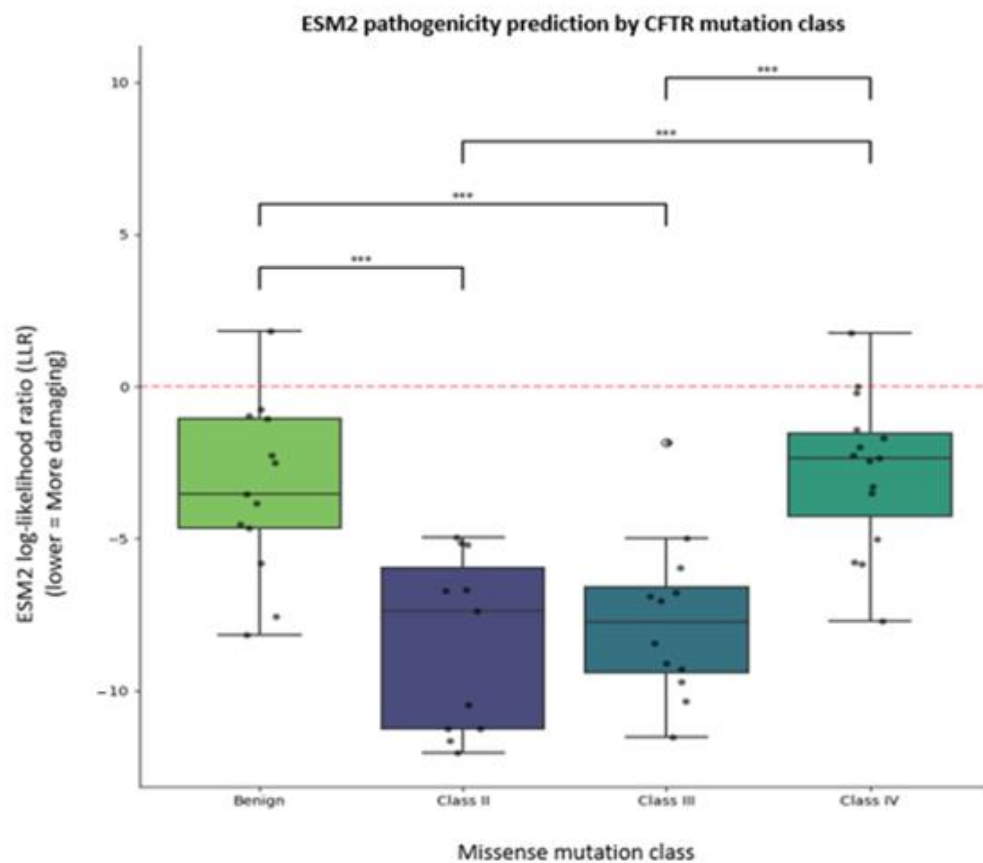


Figure 2. ESM2 pathogenicity prediction by CFTR mutation class. The pathogenicity was calculated as the log-likelihood ratio of the WT vs mutant probability. Dashed red line represents the neutral line where the WT and Mutant have equal probability. Significant ($p < 0.001$) differences are represented as ***.

The results demonstrate that ESM-2 effectively distinguishes benign polymorphisms from pathogenic CFTR missense variants. Benign variants exhibited significantly higher (less negative) log-likelihood ratio (LLR) scores (mean = -3.37), indicating that the model recognises these substitutions as evolutionarily compatible with the protein's sequence constraints. In contrast, Class II (misfolding/trafficking) and Class III (gating) mutations yielded significantly lower LLR scores ($p < 0.001$ compared to benign), consistent with the fact that these variants disrupt highly conserved structural or functional motifs. The lowest mean LLR for Class II (-8.42) reflects the severe instability caused by substitutions at tightly constrained residues, while Class III mutations were similarly penalised due to their impact on the channel's critical gating machinery (mean = -7.65). Notably, Class IV mutations displayed intermediate LLR values (mean = -2.77): the model interpreted these substitutions as moderately disruptive but significantly less damaging than the structural failures of Class II and III. This aligns perfectly with the established clinical severity hierarchy, where Class IV

mutations are less severe than Class II and Class III (Brodie *et al.*, 2015; Lopes-Pacheco, 2020; Regard *et al.*, 2023). Class IV mutations are known to retain residual channel function and conductance, a biological subtlety successfully captured by the model's probability scoring.

Conclusion

Bioinformatics has fundamentally shifted the scientific process by transforming raw data into actionable experimental roadmaps. Instead of relying on trial-and-error, researchers now use computational predictions to prioritise specific targets for wet-lab validation, effectively guiding the direction of new experiments. Conversely, bioinformatics consolidates these scattered biological findings into unified systems-level frameworks—such as Gene Ontology networks and interactome maps—allowing scientists to visualise how isolated molecular components interact to drive the complex machinery of human disease.

The emergence of deep learning, and especially PLMs, has expanded this capability dramatically. By learning the “grammar” of amino-acid sequences from millions of proteins, PLMs can identify which substitutions are structurally plausible and which are likely to disrupt protein function. In the case of CFTR, this allowed us to predict the pathogenicity of missense mutations directly from sequence context, offering insights into folding defects, gating abnormalities, and conductance impairment—ultimately *revealing how specific mutations may cause or contribute to disease*. An interesting extension to this investigation will be to analyse *unknown* missense variants and determine the pathogenicity of such mutations.

Nevertheless, it is important to recognise that bioinformatics *generates hypotheses, not final answers*. Wet-lab approaches therefore remain essential counterparts to in silico methods, ensuring that computational insights translate into genuine mechanistic understanding and ultimately into clinical impact.

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